

Mark Scheme

Q1.

Question	Scheme	Marks	AOs
(a)	Attempts $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$	M1	1.1b
	$\frac{dy}{dx} = \frac{\sqrt{3} \sin 2t}{\sin t} \quad (= 2\sqrt{3} \cos t)$	A1	1.1b
	(2)		
(b)	Substitutes $t = \frac{2\pi}{3}$ in $\frac{dy}{dx} = \frac{\sqrt{3} \sin 2t}{\sin t} = (-\sqrt{3})$	M1	2.1
	Uses gradient of normal $= -\frac{1}{dy/dx} = \left(\frac{1}{\sqrt{3}}\right)$	M1	2.1
	Coordinates of $P = \left(-1, -\frac{\sqrt{3}}{2}\right)$	B1	1.1b
	Correct form of normal $y + \frac{\sqrt{3}}{2} = \frac{1}{\sqrt{3}}(x+1)$	M1	2.1
	Completes proof $\Rightarrow 2x - 2\sqrt{3}y - 1 = 0$ *	A1*	1.1b
	(5)		
(c)	Substitutes $x = 2\cos t$ and $y = \sqrt{3} \cos 2t$ into $2x - 2\sqrt{3}y - 1 = 0$	M1	3.1a
	Uses the identity $\cos 2t = 2\cos^2 t - 1$ to produce a quadratic in $\cos t$	M1	3.1a
	$\Rightarrow 12\cos^2 t - 4\cos t - 5 = 0$	A1	1.1b
	Finds $\cos t = \frac{5}{6}, \frac{1}{2}$	M1	2.4
	Substitutes their $\cos t = \frac{5}{6}$ into $x = 2\cos t, y = \sqrt{3} \cos 2t,$	M1	1.1b
	$Q = \left(\frac{5}{3}, \frac{7}{18}\sqrt{3}\right)$	A1	1.1b
	(6)		
(13 marks)			

Notes:

(a)

M1: Attempts $\frac{dy}{dx} = \frac{dy}{dx} \frac{dt}{dt}$ and achieves a form $k \frac{\sin 2t}{\sin t}$ Alternatively candidates may apply the double angle identity for $\cos 2t$ and achieve a form $k \frac{\sin t \cos t}{\sin t}$

A1: Scored for a correct answer, either $\frac{\sqrt{3} \sin 2t}{\sin t}$ or $2\sqrt{3} \cos t$

(b)

M1: For substituting $t = \frac{2\pi}{3}$ in their $\frac{dy}{dx}$ which must be in terms of t

M1: Uses the gradient of the normal is the negative reciprocal of the value of $\frac{dy}{dx}$. This may be seen in the equation of l .

B1: States or uses (in their tangent or normal) that $P = \left(-1, -\frac{\sqrt{3}}{2}\right)$

M1: Uses their numerical value of $-1/\frac{dy}{dx}$ with their $\left(-1, -\frac{\sqrt{3}}{2}\right)$ to form an equation of the normal at P

A1*: This is a proof and all aspects need to be correct. Correct answer only $2x - 2\sqrt{3}y - 1 = 0$

(c)

M1: For substituting $x = 2\cos t$ and $y = \sqrt{3} \cos 2t$ into $2x - 2\sqrt{3}y - 1 = 0$ to produce an equation in t . Alternatively candidates could use $\cos 2t = 2\cos^2 t - 1$ to set up an equation of the form $y = Ax^2 + B$.

M1: Uses the identity $\cos 2t = 2\cos^2 t - 1$ to produce a quadratic equation in $\cos t$
In the alternative method it is for combining their $y = Ax^2 + B$ with $2x - 2\sqrt{3}y - 1 = 0$ to get an equation in just one variable

A1: For the correct quadratic equation $12\cos^2 t - 4\cos t - 5 = 0$
Alternatively the equations in x and y are $3x^2 - 2x - 5 = 0$ $12\sqrt{3}y^2 + 4y - 7\sqrt{3} = 0$

M1: Solves the quadratic equation in $\cos t$ (or x or y) and rejects the value corresponding to P .

M1: Substitutes their $\cos t = \frac{5}{6}$ or their $t = \arccos\left(\frac{5}{6}\right)$ in $x = 2\cos t$ and $y = \sqrt{3} \cos 2t$

If a value of x or y has been found it is for finding the other coordinate.

A1: $Q = \left(\frac{5}{3}, \frac{7}{18}\sqrt{3}\right)$. Allow $x = \frac{5}{3}, y = \frac{7}{18}\sqrt{3}$ but do not allow decimal equivalents.

Q2.

Question	Scheme	Marks	AOs
(a)	Attempts $y \cdot \frac{dx}{dt} = (2 \sin 2t + 3 \sin t) \times 16 \sin t \cos t$ and uses $\sin 2t = 2 \sin t \cos t$	M1	2.1
	Correct expanded integrand. Usually for one of $(R) = \int \underline{48 \sin^2 t \cos t + 16 \sin^2 2t} dt$ $(R) = \int \underline{48 \sin^2 t \cos t + 64 \sin^2 t \cos^2 t} dt$ $(R) = \int \underline{24 \sin 2t \sin t + 16 \sin^2 2t} dt$	A1	1.1b
	Attempts to use $\cos 4t = 1 - 2 \sin^2 2t = (1 - 8 \sin^2 t \cos^2 t)$	M1	1.1b
	$R = \int_0^a 8 - 8 \cos 4t + 48 \sin^2 t \cos t dt$ *	A1*	2.1
	Deduces $a = \frac{\pi}{4}$	B1	2.2a
		(5)	
(b)	$\int 8 - 8 \cos 4t + 48 \sin^2 t \cos t dt = 8t - 2 \sin 4t + 16 \sin^3 t$	M1 A1	2.1 1.1b
	$\left[8t - 2 \sin 4t + 16 \sin^3 t \right]_0^{\frac{\pi}{4}} = 2\pi + 4\sqrt{2}$	M1 A1	2.1 1.1b
		(4)	
(9 marks)			
Notes:			

(a) Condone work in another variable, say $\theta \leftrightarrow t$ if used consistently for the first 3 marks

M1: For the key step in attempting $y \cdot \frac{dx}{dt} = (2 \sin 2t + 3 \sin t) \times 16 \sin t \cos t$ with an attempt to use

$\sin 2t = 2 \sin t \cos t$ Condone slips in finding $\frac{dx}{dt}$ but it must be of the form $k \sin t \cos t$

E.g. I $y \cdot \frac{dx}{dt} = (2 \sin 2t + 3 \sin t) \times k \sin t \cos t = (4 \sin t \cos t + 3 \sin t) \times k \sin t \cos t$

E.g. II $y \cdot \frac{dx}{dt} = (2 \sin 2t + 3 \sin t) \times k \sin t \cos t = (2 \sin 2t + 3 \sin t) \times \frac{k}{2} \sin 2t$

A1: A correct (expanded) integrand in t . Don't be concerned by the absence of $\int$ or dt or limits

$(R) = \int \underline{48 \sin^2 t \cos t + 16 \sin^2 2t} dt$ or $(R) = \int \underline{48 \sin^2 t \cos t + 64 \sin^2 t \cos^2 t} dt$

but watch for other correct versions such as $(R) = \int \underline{24 \sin 2t \sin t + 16 \sin^2 2t} dt$

M1: Attempts to use $\cos 4t = \pm 1 \pm 2 \sin^2 2t$ to get the integrand in the correct form.

If they have the form $P \sin^2 2t$ it is acceptable to write $P \sin^2 2t = \frac{P}{2}(\pm 1 \pm \cos 4t)$

If they have the form $Q \sin^2 t \cos^2 t$ sight and use of $\sin 2t$ and/or $\cos 2t$ will usually be seen first.

There are many ways to do this, below is such an example

$$Q \sin^2 t \cos^2 t = Q \left(\frac{1 - \cos 2t}{2} \right) \left(\frac{1 + \cos 2t}{2} \right) = Q \left(\frac{1 - \cos^2 2t}{4} \right) = Q \left(\frac{1}{4} - \frac{\cos^2 2t}{4} \right) = Q \left(\frac{1}{4} - \frac{1 + \cos 4t}{8} \right)$$

Allow candidates to start with the given answer and work backwards using the same rules.

So expect to see $\cos 4t = \pm 1 \pm 2 \sin^2 2t$ or $\cos 4t = \pm 2 \cos^2 2t \pm 1$ before double angle identities for $\sin 2t$ or $\cos 2t$ are used.

A1*: Proceeds to the given answer with correct working. The order of the terms is not important.

Ignore limits for this mark. The integration sign and the dt must be seen on their final answer.

If they have worked backwards there must be a concluding statement to the effect that they know that they have shown it. The integration sign and the dt must also be seen

E.g. Reaches $\int 48 \sin^2 t \cos t + 64 \sin^2 t \cos^2 t \, dt$

$$\begin{aligned} \text{Answer is } & \int 8 - 8 \cos 4t + 48 \sin^2 t \cos t \, dt \\ &= \int 8 - 8(1 - 2 \sin^2 2t) + 48 \sin^2 t \cos t \, dt \\ &= \int 16 \sin^2 2t + 48 \sin^2 t \cos t \, dt \\ &= \int 64 \sin^2 t \cos^2 t + 48 \sin^2 t \cos t \, dt \end{aligned}$$

which is the same, ✓

B1: Deduces $a = \frac{\pi}{4}$. It may be awarded from the upper limit and can be awarded from (b)

(b)

M1: For the key process in using a correct approach to integrating the trigonometric terms.

May be done separately.

There may be lots of intermediate steps (e.g. let $u = \sin t$).

There are other more complicated methods so look carefully at what they are doing.

$$\int 8 - 8 \cos 4t + 48 \sin^2 t \cos t \, dt = \dots \pm P \sin 4t \pm Q \sin^3 t \text{ where } P \text{ and } Q \text{ are constants}$$

A1: $\int 8 - 8 \cos 4t + 48 \sin^2 t \cos t \, dt = 8t - 2 \sin 4t + 16 \sin^3 t (+c)$

If they have written $16 \sin^3 t$ as $16 \sin^3$ only award if further work implies a correct answer.

Similarly, $8t$ may be written as $8x$. Award if further work implies $8t$, e.g. substituting in their limits.

Do not penalise this sort of slip at all, these are intermediate answers.

M1: Uses the limits their a and 0 where $a = \frac{\pi}{6}, \frac{\pi}{4}$ or $\frac{\pi}{3}$ in an expression of the form $kt \pm P \sin 4t \pm Q \sin^3 t$ leading to an exact answer. Ignore evidence at lower limit as terms are 0

A1: CSO $2\pi + 4\sqrt{2}$ or exact simplified equivalent such as $2\pi + \frac{8}{\sqrt{2}}$ or $2\pi + \sqrt{32}$.

Be aware that $\int_0^{\frac{\pi}{4}} 8 - 8 \cos 4t + 48 \sin^2 t \cos t \, dt = 8t + \lambda \sin 4t + 16 \sin^3 t (+c)$ would lead to the correct answer but would only score M1 A0 M1 A0

Q3.

Question	Scheme	Marks	AOs
	$(x-3)^2 + y^2 = \left(\frac{t^2+5}{t^2+1} - 3\right)^2 + \left(\frac{4t}{t^2+1}\right)^2$	M1	3.1a
	$= \frac{(2-2t^2)^2 + 16t^2}{(t^2+1)^2} = \frac{4+8t^2+4t^4}{(t^2+1)^2}$	dM1	1.1b
	$\frac{4(t^4+2t^2+1)}{(t^2+1)^2} = \frac{4(t^2+1)^2}{(t^2+1)^2} = 4^*$	A1*	2.1
	(3)		

M1: Attempts to substitute the given parametric forms into the Cartesian equation or the lhs of the Cartesian equation. There may have been an (incorrect) attempt to multiply out the $(x-3)^2$ term.
 dM1: Attempts to combine (at least the lhs) using correct processing into a single fraction, multiplies out and collects terms on the numerator.
 A1*: Fully correct proof showing all key steps

Question	Scheme	Marks	AOs
Alt	$x = \frac{t^2+5}{t^2+1} \Rightarrow xt^2 + x = t^2 + 5 \Rightarrow t^2 = \frac{5-x}{x-1}$ $y = \frac{4t}{t^2+1} \Rightarrow y^2 = \frac{16t^2}{(t^2+1)^2} = \frac{16\left(\frac{5-x}{x-1}\right)}{\left(\frac{5-x}{x-1} + 1\right)^2}$	M1	3.1a
	$y^2 = \frac{16\left(\frac{5-x}{x-1}\right)}{\left(\frac{5-x}{x-1} + 1\right)^2} = 16\left(\frac{5-x}{x-1}\right) \times \left(\frac{(x-1)}{5-x+x-1}\right)^2 \Rightarrow y^2 = (5-x)(x-1)$	dM1	1.1b
	$y^2 = (5-x)(x-1) \Rightarrow y^2 = 6x - x^2 - 5$ $\Rightarrow y^2 = 4 - (x-3)^2 \text{ or other intermediate step}$ $\Rightarrow (x-3)^2 + y^2 = 4^*$	A1*	2.1
	(3)		
(3 marks)			
Notes			

M1: Adopts a correct strategy for eliminating t to obtain an equation in terms of x and y only. See scheme.
 Other methods exist which also lead to an appropriate equation. E.g using $t = \frac{y}{x-1}$
 dM1: Uses correct processing to eliminate the fractions and start to simplify
 A1*: Fully correct proof showing all key steps

Q4.

Part	Working or answer an examiner might expect to see	Mark	Notes
	$(10 \cos t)^2 + (4\sqrt{2} \sin t)^2 = 66$	M1	This mark is given for combining the two equations to show where the curve and circle meet
	$100 (\cos t)^2 + 32(1 - \cos t)^2 = 66$	M1	This mark is given for forming an equation in $\cos t$ only
	$68 \cos^2 t = 34$	A1	This mark is given for simplifying to find an equation in terms of $\cos t$
	$\cos t = \pm \frac{1}{\sqrt{2}} \Rightarrow t = \frac{\pi}{4}$	M1	This mark is given for finding a value for t
	$x = 10 \times \frac{1}{\sqrt{2}}$ $y = 4\sqrt{2} \times -\sin \frac{\pi}{4} = 4\sqrt{2} \times -\frac{1}{\sqrt{2}}$	M1	This mark is given for a method to substitute back into the original equations to find value for x and y
	$S = (5\sqrt{2}, -4)$	A1	This mark is given for the correct coordinates of S
(Total 6 marks)			

Q5.

Question	Scheme	Marks	AOs
(a)(i)	$y \times \frac{dx}{dt} = 5 \sin 2t \times 6 \cos t \text{ or } 5 \times 2 \sin t \cos t \times 6 \cos t$	M1	1.2

	$(\text{Area} =) \int 5 \sin 2t \times 6 \cos t \, dt = \int 5 \times 2 \sin t \cos t \times 6 \cos t \, dt$ or $\int 5 \sin 2t \times 6 \cos t \, dt = \int 60 \sin t \cos^2 t \, dt$	dM1	1.1b
	$(\text{Area} =) \int_0^{\frac{\pi}{2}} 60 \sin t \cos^2 t \, dt *$	A1*	2.1*
		(3)	
(a)(ii)	$\int 60 \sin t \cos^2 t \, dt = -20 \cos^3 t$	M1 A1	1.1b 1.1b
	$\text{Area} = [-20 \cos^3 t]_0^{\frac{\pi}{2}} = 0 - (-20) = 20 *$	A1*	2.1
		(3)	
(b)	$5 \sin 2t = 4.2 \Rightarrow \sin 2t = \frac{4.2}{5}$	M1	3.4
	$t = 0.4986..., 1.072...$	A1	1.1b
	Attempts to find the x values at both t values	dM1	3.4
	$t = 0.4986... \Rightarrow x = 2.869...$ $t = 1.072 \Rightarrow x = 5.269...$	A1	1.1b
	Width of path = 2.40 metres	A1	3.2a
		(5)	
(11 marks)			

Notes:

(a)(i)

M1: Attempts to multiply y by $\frac{dx}{dt}$ to obtain $A \sin 2t \cos t$ but may apply $\sin 2t = 2 \sin t \cos t$ here

dM1: Attempts to use $\sin 2t = 2 \sin t \cos t$ within an integral which may be implied by

e.g. $A \int \sin 2t \times \cos t \, dt = \int k \sin t \cos^2 t \, dt$

A1*: Fully correct work leading to the given answer.

This must include $\sin 2t = 2 \sin t \cos t$ or e.g. $5 \sin 2t = 10 \sin t \cos t$ seen explicitly in their proof and a correct intermediate line that includes an integral sign and the “dt”

Allow the limits to just “appear” in the final answer e.g. working need not be shown for the limits.

(a)(ii)

M1: Obtains $\int 60 \sin t \cos^2 t \, dt = k \cos^3 t$. This may be attempted via a substitution of $u = \cos t$ to obtain

$$\int 60 \sin t \cos^2 t \, dt = k u^3$$

A1: Correct integration $-20 \cos^3 t$ or equivalent e.g. $-20u^3$

A1*: Rigorous proof with all aspects correct including the correct limits and the $0 - (-20)$ and

not just: $-20 \cos^3 \frac{\pi}{2} - (-20 \cos^3 0) = 20$

(b)

M1: Uses the given model and attempts to find value(s) of t when $\sin 2t = \frac{4.2}{5}$. Look for $2t = \sin^{-1} \frac{4.2}{5} \Rightarrow t = \dots$

A1: At least one correct value for t, correct to 2 dp. FYI $t = 0.4986..., 1.072...$ or in degrees $t = 28.57..., 61.42...$

dM1: Attempts to find **TWO** distinct values of x when $\sin 2t = \frac{4.2}{5}$. Condone poor trig work and allow this mark if 2

values of x are attempted from 2 values of t .

A1: Both values correct to 2 dp. NB $x = 2.869\dots, 5.269\dots$

Or may take Cartesian approach

$$5 \sin 2t = 4.2 \Rightarrow 10 \sin t \cos t = 4.2 \Rightarrow 10 \frac{x}{6} \sqrt{1 - \frac{x^2}{36}} = 4.2 \Rightarrow x^4 - 36x^2 + 228.6144 = 0 \Rightarrow x = 2.869\dots, 5.269\dots$$

M1: For converting to Cartesian form A1: Correct quartic M1: Solves quartic A1: Correct values

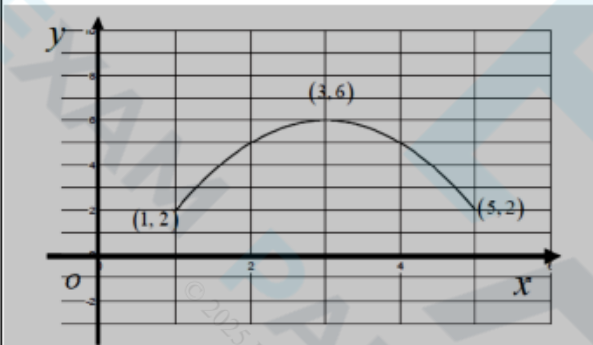
A1: 2.40 metres or 240 cm

Allow awrt 2.40 m or allow 2.4m (not awrt 2.4 m) and allow awrt 240 cm. Units are required.

Q6.

Question	Scheme	Marks	AOs
	Attempts to substitute $= \frac{x+1}{2}$ into $y \Rightarrow y = 4\left(\frac{x+1}{2}\right) - 7 + \frac{6}{(x+1)}$	M1	2.1
	Attempts to write as a single fraction $y = \frac{(2x-5)(x+1)+6}{(x+1)}$	M1	2.1
	$y = \frac{2x^2-3x+1}{x+1} \quad a=-3, b=1$	A1	1.1b
(3 marks)			
Notes:			
M1: Score for an attempt at substituting $t = \frac{x+1}{2}$ or equivalent into $y = 4t - 7 + \frac{3}{t}$			
M1: Award this for an attempt at a single fraction with a correct common denominator. Their $4\left(\frac{x+1}{2}\right) - 7$ term may be simplified first			
A1: Correct answer only $y = \frac{2x^2-3x+1}{x+1} \quad a=-3, b=1$			

Q7.

Question	Scheme	Marks	AOs
(a)	Attempts to use $\cos 2t = 1 - 2\sin^2 t \Rightarrow \frac{y-4}{2} = 1 - 2\left(\frac{x-3}{2}\right)^2$	M1	2.1
	$\Rightarrow y - 4 = 2 - 4 \times \frac{(x-3)^2}{4} \Rightarrow y = 6 - (x-3)^2 *$	A1*	1.1b
		(2)	
(b)	 shaped parabola Fully correct with 'ends' at (1,2) & (5,2) Suitable reason : Eg states as $x = 3 + 2\sin t, 1 \leq x \leq 5$	M1	1.1b
		A1	1.1b
		B1	2.4
		(3)	

(c)	Either finds the lower value for $k = 7$ or deduces that $k < \frac{37}{4}$	B1	2.2a
	Finds where $x + y = k$ meets $y = 6 - (x-3)^2$ $\Rightarrow k - x = 6 - (x-3)^2$ and proceeds to 3TQ in x or y	M1	3.1a
	Correct 3TQ in x $x^2 - 7x + (k+3) = 0$ Or y $y^2 + (7-2k)y + (k^2 - 6k + 3) = 0$	A1	1.1b
	Uses $b^2 - 4ac = 0 \Rightarrow 49 - 4 \times 1 \times (k+3) = 0 \Rightarrow k = \left(\frac{37}{4}\right)$ or $(7-2k)^2 - 4 \times 1 \times (k^2 - 6k + 3) = 0 \Rightarrow k = \left(\frac{37}{4}\right)$	M1	2.1
	Range of values for $k = \left\{k : 7 \leq k < \frac{37}{4}\right\}$	A1	2.5
		(5)	
(10 marks)			

(a)
M1: Uses $\cos 2t = 1 - 2\sin^2 t$ in an attempt to eliminate t
A1*: Proceeds to $y = 6 - (x - 3)^2$ without any errors
Allow a proof where they start with $y = 6 - (x - 3)^2$ and substitute the parametric coordinates. M1 is scored for a correct $\cos 2t = 1 - 2\sin^2 t$ but A1 is only scored when both sides are seen to be the same AND a comment is made, hence proven, or similar.
(b)
M1: For sketching a $\cap$ parabola with a maximum in quadrant one. It does not need to be symmetrical
A1: For sketching a $\cap$ parabola with a maximum in quadrant one and with end coordinates of $(1, 2)$ and $(5, 2)$
B1: Any suitable explanation as to why C does not include all points of $y = 6 - (x - 3)^2$, $x \in \mathbb{R}$ This should include a reference to the limits on sin or cos with a link to a restriction on x or y . For example 'As $-1 \leq \sin t \leq 1$ then $1 \leq x \leq 5$ ' Condone in words ' x lies between 1 and 5' and strict inequalities 'As $\sin t \leq 1$ then $x \leq 5$ ' Condone in words ' x is less than 5' 'As $-1 \leq \cos(2t) \leq 1$ then $2 \leq y \leq 6$ ' Condone in words ' y lies between 2 and 6' Withhold if the statement is incorrect Eg "because the domain is $2 \leq x \leq 5$ " Do not allow a statement on the top limit of y as this is the same for both curves
(c)
B1: Deduces either
- the correct that the lower value of $k = 7$ This can be found by substituting into $(5, 2)$ $x + y = k \Rightarrow k = 7$ or substituting $x = 5$ into $x^2 - 7x + (k + 3) = 0 \Rightarrow 25 - 35 + k + 3 = 0 \Rightarrow k = 7$ - or deduces that $k < \frac{37}{4}$ This may be awarded from later work
M1: For an attempt at the upper value for k .
Finds where $x + y = k$ meets $y = 6 - (x - 3)^2$ once by using an appropriate method.
Eg. Sets $k - x = 6 - (x - 3)^2$ and proceeds to a 3TQ
A1: Correct 3TQ $x^2 - 7x + (k + 3) = 0$ The $= 0$ may be implied by subsequent work
M1: Uses the "discriminant" condition. Accept use of $b^2 = 4ac$ or $b^2 \leq 4ac$ where ... is any inequality leading to a critical value for k . Eg. one root $\Rightarrow 49 - 4 \times 1 \times (k + 3) = 0 \Rightarrow k = \frac{37}{4}$
A1: Range of values for $k = \left\{ k : 7 \leq k < \frac{37}{4} \right\}$ Accept $k \in \left[7, \frac{37}{4} \right)$ or exact equivalent

ALT	As above	B1	2.2a
	Finds where $x + y = k$ meets $y = 6 - (x - 3)^2$ once by using an appropriate method. Eg. Sets gradient of $y = 6 - (x - 3)^2$ equal to -1	M1	3.1a
	$-2x + 6 = -1 \Rightarrow x = 3.5$	A1	1.1b
	Finds point of intersection and uses this to find upper value of k . $y = 6 - (3.5 - 3)^2 = 5.75$ Hence using $k = 3.5 + 5.75 = 9.25$	M1	2.1
	Range of values for $k = \{k : 7 \leq k < 9.25\}$	A1	2.5